A SEQUENCE WITH CONSECUTIVE RATIOS AT LEAST $1 + c/\log n$ WHOSE IRRATIONAL DILATES ARE DENSE MODULO ONE, AND THE STATUS OF A FORMAL STATEMENT OF BUGEAUD'S PROBLEM 10.6

MATH0

Abstract. We construct a strictly increasing sequence $(m_n)_{n\geq 0}$ of positive integers with $m_{n+1}/m_n \geq 1 + 1/(8\log n)$ for every $n \geq 2$ such that $(\xi m_n)_{n\geq 0}$ is dense modulo one for every irrational ξ. This proves the statement `Bugeaud06.problem_10_6_variant_1` of the formal-conjectures repository, which reads Bugeaud's Problem 10.6 ("to find a very rapidly increasing sequence" with this density property) as a ratio floor $1 + c/\log n$ and labels it as an open research problem. The statement is, however, a special case of a 1983 theorem of Ajtai, Havas and Komlós that Bugeaud records in the same book, so it does not express the open problem; this is a misformalization report, not progress on Problem 10.6, and we propose a corrected labelling and an accompanying solved statement. Our proof is elementary and self-contained and gives density only, not uniform distribution; it is verified in Lean 4 against the upstream formal-conjectures statement.

1. Introduction

For a sequence (m_n) of positive integers and $\xi \in \mathbb{R}$ one asks whether the fractional parts of ξm_n are dense in $[0, 1)$, that is, whether the image of (ξm_n) in $\mathbb{R}/\mathbb{Z}$ is dense. For lacunary sequences ($m_{n+1}/m_n \geq \lambda > 1$ for all n) the answer is negative for some irrational ξ: answering a question of Erdős, Pollington [10] and de Mathan [7] showed that such ξ always exist, and indeed there are uncountably many of them [3, pp. 29 and 215]. On the other hand, Furstenberg [5] proved that $(\xi\, 2^a 3^b)_{a,b\geq 1}$ is dense modulo one for every irrational ξ. Between these two facts Bugeaud poses the following problem [3, Problem 10.6, p. 215]:

> *To find a very rapidly increasing sequence $(m_n)_{n\geq 1}$ of positive integers such that $(\xi m_n)_{n\geq 1}$ is dense modulo one for every irrational number ξ.*

A known theorem. In the notes to Chapter 2 of the same book [3, §2.8, p. 45], Bugeaud records that "Ajtai, Havas and Komlós [26] showed that, for any given sequence $(\varepsilon_n)_{n\geq 1}$ of positive real numbers tending to 0, there exists a sequence $(m_n)_{n\geq 1}$ of positive integers such that $m_{n+1}/m_n > 1 + \varepsilon_n$ for $n \geq 1$ and the sequence $(\xi m_n)_{n\geq 1}$ is uniformly distributed modulo one for every irrational number ξ; see also [121]", where [26] and [121] are [1] and [2]. We rely on this statement of the theorem of [1] as recorded by Bugeaud. Gao, Mooroogen and Yip [6, Example 2.4(d)] describe the same result in the form: if (b_n) has sub-exponential growth and $d_n \to \infty$, there is (a_n) with $b_n \leq a_n \leq b_n + d_n$ such that (xa_n) is uniformly distributed modulo one for every real $x \neq 0$.

The formal statement. The formal-conjectures repository [4] formalizes Problem 10.6 in `FormalConjectures/Books/BugeaudDistributionModuloOne/Problem10_6.lean`. Its module docstring lists three readings of "very rapidly increasing" (a lower bound on the terms, an upper bound on the counting function, a lower bound on the ratios), records that the first two are not open, and states that the third, the ratio floor $m_{n+1}/m_n \geq 1 + c/\log n$, "remains

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open". That reading is `Bugeaud06.problem_10_6_variant_1`, tagged `research open`. With $\varepsilon_n = 1/\log(n+2)$, the theorem quoted above implies it directly (Section 4), so the label is incorrect; and since Bugeaud records that theorem in the same book, this reading is unlikely to be what Problem 10.6 asks.

Contribution and status. We give an elementary, self-contained proof of `problem_10_6_variant_1` (Theorem 1) and verify it in Lean 4. The proof does not rely on [1], and it yields only density, not uniform distribution. Since the formal statement is a special case of a known theorem and does not capture the open problem, we classify this result as a *misformalization report*; it is not progress on Problem 10.6 under any reading that is still open. If the upstream reading is instead taken at face value as the intended mathematics, what remains is a formal proof of a special case of a known result, not a new theorem. Section 4 explains the defect and proposes a corrected statement. We found no earlier formal proof: `problem_10_6_variant_1` is still proved by `sorry` in the formal-conjectures repository and in the fork that supplies `formal_proof` links for other statements of the same file (a complete search of public Lean code was not possible).

2. Statement

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, log is the natural logarithm, and a sequence (t_n) of reals is *dense modulo one* if its image under the quotient map $\pi\colon \mathbb{R} \to \mathbb{R}/\mathbb{Z}$ is dense. Following [4], call $m\colon \mathbb{N} \to \mathbb{N}$ *genuinely sublacunary* if there is $c > 0$ such that

$$1 + \frac{c}{\log n} \le \frac{m(n+1)}{m(n)} \qquad \text{for all sufficiently large } n. \tag{1}$$

(In Lean, $\log 0 = 0$ and division by zero gives 0; this plays no role, since (1) is only required for large n and our sequence has positive terms.)

Theorem 1. *There is a strictly increasing $m\colon \mathbb{N} \to \mathbb{N}$ with $m(n) \ge 1$ for all n and*

$$m(n+1) \ge \Big(1 + \frac{1}{8\log n}\Big) m(n) \qquad \text{for all } n \ge 2,$$

such that $(\xi m(n))_{n\in\mathbb{N}}$ is dense modulo one for every irrational ξ. In particular m is genuinely sublacunary, with $c = 1/8$.

The upstream statement, copied verbatim from `Problem10_6.lean` at commit `c77fdd4` (inside `namespace Bugeaud06`, with `open Filter`; the docstring of the theorem is omitted), is:

```
/-- The sequence $m$ eventually grows at least geometrically with a logarithmic correction. -/
def IsGenuinelySublacunary (m : ℕ → ℕ) : Prop :=
  ∃ c > 0, ∀ᶠ (n : ℕ) in atTop, (1 + c / Real.log n) ≤ (m (n+1) : ℝ) / m n

@[category research open, AMS 11]
theorem problem_10_6_variant_1 :
    ∃ m : ℕ → ℕ,
    StrictMono m ∧
    IsGenuinelySublacunary m ∧
    ∀ ξ : ℝ, Irrational ξ →
      Dense (Set.range fun n => (↑(ξ * m n) : AddCircle (1 : ℝ))) := by
  sorry
```

Here `AddCircle (1 : ℝ)` is $\mathbb{R}/\mathbb{Z}$ with its quotient topology, and `↑` is the quotient map π. The omitted docstring states Problem 10.6, says that this ratio reading "is the one that remains open", and argues that it "is not answered by runs of consecutive integers, since inside a run $m_{n+1} - m_n = 1$". Theorem 1 proves this statement. The formal statement indexes the sequence from 0 rather than from 1 as in [3], which is immaterial.

3. Proof

3.1. **The construction.** For $k \in \mathbb{N}$ put

$$K_k = 2^k, \quad M_k = 2^{4^k}, \quad D_k = 2^{M_k}, \quad W_k = 2D_kK_k, \quad A_k = K_kW_k, \quad L_k = M_k + 2k + 1,$$
$$B_k = \{A_k + rW_k + (D_k + r)j : 0 \le r < K_k,\ 0 \le j < K_k\}, \qquad S = \{2^i : i \in \mathbb{N}\} \cup \bigcup_{k\in\mathbb{N}} B_k.$$

Thus B_k consists of K_k arithmetic progressions of length K_k with the slightly different large differences $D_k + r$, the r-th one starting at $A_k + rW_k$. The powers of two are included so that every element of S has a large index (Lemma 4). When k is clear we drop the subscript.

Lemma 2 (Block arithmetic)**.** *Let $k \in \mathbb{N}$ and $r, j < K$.*

(i) $K \le D$.
(ii) $A = 2^L$ *and* $2A = 4 \cdot 4^k D$.
(iii) $(D + r)j + D \le W$.
(iv) *Every $x \in B_k$ satisfies $A \le x$ and $x + D \le 2A$.*
(v) *If $x, y \in B_k$ and $x < y$, then $x + D \le y$.*
(vi) *If $k < k'$, then $L_k + 1 \le L_{k'}$.*

Proof. (i) $k \le 2^k \le 4^k \le 2^{4^k} = M$, so $K = 2^k \le 2^M = D$. (ii) $A = 2DK^2 = 2 \cdot 2^M \cdot 4^k = 2^{M+2k+1}$, and $2A = 4DK^2 = 4 \cdot 4^k D$. (iii) Using $r < K$, $j + 1 \le K$ and (i): $(D + r)j + D \le (D + r)(j + 1) \le (D + K)K \le 2DK = W$. (iv) Let $x = A + rW + (D + r)j$. Clearly $x \ge A$, and by (iii), $x + D \le A + (r + 1)W \le A + KW = 2A$. (v) Write $x = A + rW + (D + r)j$ and $y = A + r'W + (D + r')j'$. If $r < r'$ then $y \ge A + (r + 1)W \ge x + D$ by (iii). If $r = r'$ then $j < j'$ and $y - x = (D + r)(j' - j) \ge D$. If $r > r'$ then $x \ge A + (r' + 1)W \ge y + D$ by (iii), contradicting $x < y$. (vi) $M_k \le M_{k'}$ and $2k + 1 < 2k' + 1$. □

Lemma 3 (Gaps)**.** *Let $x, y \in S$ with $x < y$.*

(a) *If $x \in B_k$, then $y \ge x + D_k$.*
(b) *If $x = 2^i$ with $i \ne L_k$ for every k, then $y \ge 2x$.*

Moreover, every $x \in S$ falls under (a) or (b).

Proof. (a) Suppose first $y = 2^{i'}$. If $i' \le L_k$ then $y \le 2^{L_k} = A_k \le x$ by Lemma 2(ii),(iv), which is impossible; so $i' \ge L_k + 1$ and $y \ge 2A_k \ge x + D_k$. Suppose next $y \in B_{k'}$. If $k' = k$, use Lemma 2(v). If $k < k'$, then by Lemma 2(vi), $y \ge A_{k'} = 2^{L_{k'}} \ge 2^{L_k+1} = 2A_k \ge x + D_k$. If $k' < k$, then $y < 2A_{k'} = 2^{L_{k'}+1} \le 2^{L_k} = A_k \le x$, which is impossible.

(b) If $y = 2^{i'}$ then $i' > i$ and $y \ge 2^{i+1}$. If $y \in B_k$ then $2^i < y < 2A_k = 2^{L_k+1}$, so $i \le L_k$; as $i \ne L_k$, we get $i + 1 \le L_k$ and $y \ge A_k = 2^{L_k} \ge 2^{i+1}$.

Finally, a power 2^{L_k} equals $A_k \in B_k$ (take $r = j = 0$), so it falls under (a). □

Lemma 4 (Enumeration)**.** *The set S is infinite. Let $m\colon \mathbb{N} \to \mathbb{N}$ be its increasing enumeration ($m(n)$ is the $(n + 1)$-st smallest element of S). Then m is strictly increasing, $m(\mathbb{N}) = S$, and $1 \le m(n) \le 2^n$ for every n.*

Proof. S contains all powers of two, so it is infinite, and the first three properties are standard facts about the enumeration of an infinite subset of $\mathbb{N}$ (in Lean, `Nat.nth` with `Nat.nth_strictMono`, `Nat.nth_mem_of_infinite` and `Nat.range_nth_of_infinite`). Every element of S is a power of two or at least $A_k \ge 1$, so $m(n) \ge 1$. For $i \in \mathbb{N}$ let $f(i)$ be the index with $m(f(i)) = 2^i$. Since m and $i \mapsto 2^i$ are strictly increasing, so is f; hence $f(n) \ge n$ and $m(n) \le m(f(n)) = 2^n$. □

Proposition 5 (Ratio floor)**.** *For every $n \ge 2$, $m(n + 1) \ge \big(1 + \frac{1}{8\log n}\big)m(n)$.*

Proof. Put $x = m(n)$ and $y = m(n+1)$, so $x < y$ and both lie in S. Since $n \geq 2$, $\log n \geq \log 2 > 1/2$. If x falls under Lemma 3(b), then $y \geq 2x \geq (1 + \frac{1}{8\log n})x$, because $\frac{1}{8\log n} < \frac{1}{4}$. Otherwise $x \in B_k$ for some k. By Lemmas 2(ii),(iv) and 4, $2^{L_k} = A_k \leq x \leq 2^n$, so $n \geq L_k \geq M_k = 2^{4^k}$ and $\log n \geq 4^k \log 2 \geq 4^k/2$. By Lemma 2(ii),(iv), $x < 2A_k = 4 \cdot 4^k D_k$. Hence

$$\frac{x}{8\log n} \leq \frac{4 \cdot 4^k D_k}{8 \cdot 4^k/2} = D_k,$$

and Lemma 3(a) gives $y \geq x + D_k \geq \big(1 + \frac{1}{8\log n}\big)x$. □

3.2. **Density.** For real t write $\lfloor t \rfloor$ for the floor and $\mathrm{frac}(t) = t - \lfloor t \rfloor \in [0, 1)$.

Lemma 6 (Stepping)**.** *Let $s \neq 0$ and $\eta > 0$ be real with $|s| \leq \eta$, and let $N \in \mathbb{N}$ with $N|s| \geq 1$. For all real u, a there are $j \in \{0, \ldots, N\}$ and $z \in \mathbb{Z}$ with $a \leq u + js + z \leq a + \eta$.*

Proof. Let $s > 0$. Put $w = \mathrm{frac}(a - u)$, $j = \lceil w/s \rceil$ and $z = \lfloor a - u \rfloor$. Then $w \leq js < w + s$, so $u + js + z = a + (js - w) \in [a, a + s) \subseteq [a, a + \eta]$. Since $w < 1 \leq Ns$, we have $w/s < N$ and $j \leq N$. If $s < 0$, apply the case just proved to $-s$, $-u$ and $-a - \eta$, obtaining $j \leq N$ and z with $-a - \eta \leq -u - js + z \leq -a$; then j and $-z$ work. □

Proposition 7. *Let ξ be irrational and $a < b$ real. There are $x \in S$ and $z \in \mathbb{Z}$ with $a < \xi x + z < b$.*

Proof. Put $\eta = (b - a)/2$ and $n_0 = \lceil 2/\eta \rceil + 1$. By Dirichlet's approximation theorem (Mathlib's `Real.exists_nat_abs_mul_sub_round_le`) there is an integer q with $1 \leq q \leq n_0$ and $|q\xi - R| \leq 1/(n_0 + 1) \leq \eta/2$, where R is the integer nearest to $q\xi$. Put $\delta = q\xi - R$. Then $\delta \neq 0$, since $q\xi$ is irrational. Let $N_1 = \lceil 1/|\delta| \rceil$, $N_2 = \lceil 2/\eta \rceil$ and $k = N_1 q + N_2$; then $K_k = 2^k > k$. Write K, D, W, A for the parameters of block k.

Step 1: choosing the progression. Lemma 6 with step δ, window length $\eta/2$ (note $|\delta| \leq \eta/2$ and $N_1|\delta| \geq 1$), $u = \xi D$, and $\eta/2$ in place of a gives $j_1 \leq N_1$ and $z_1 \in \mathbb{Z}$ such that $s := \xi D + j_1\delta + z_1 \in [\eta/2, \eta]$. Put $r = j_1 q$; then $r \leq N_1 q \leq k < K$, and

$$s = \xi(D + r) + z_1 - j_1 R.$$

Step 2: walking along it. Since $0 < s \leq \eta$ and $N_2 s \geq (2/\eta)(\eta/2) = 1$, Lemma 6 with step s, window length η, $u = \xi(A + rW)$, and $a + \eta/2$ in place of a gives $j_2 \leq N_2 < K$ and $z_2 \in \mathbb{Z}$ with $a + \eta/2 \leq \xi(A + rW) + j_2 s + z_2 \leq a + 3\eta/2$.

Let $x = A + rW + (D + r)j_2 \in B_k \subseteq S$ and $z = z_2 + j_2(z_1 - j_1 R) \in \mathbb{Z}$. By Step 1,

$$\xi x + z = \xi(A + rW) + j_2 s + z_2 \in [a + \eta/2,\ a + 3\eta/2] \subset (a, b).$$ □

Proof of Theorem 1. Let m be the enumeration of Lemma 4; it is strictly increasing with positive terms, and Proposition 5 gives the ratio floor. Let ξ be irrational. By Proposition 7, $T = \{\xi x + z : x \in S,\ z \in \mathbb{Z}\}$ is dense in $\mathbb{R}$. The map π is continuous and surjective, so $\pi(T)$ is dense in $\mathbb{R}/\mathbb{Z}$. Since $S = m(\mathbb{N})$ and $\pi(\xi x + z) = \pi(\xi x)$, we have $\pi(T) \subseteq \{\pi(\xi m(n)) : n \in \mathbb{N}\}$, which is therefore dense. □

Remark 8. The upstream docstring argues that runs of consecutive integers cannot answer the ratio reading. The construction above avoids this by replacing runs with progressions of large differences $D_k + r$; Dirichlet's theorem selects a difference $D_k + r$ whose dilate $\xi(D_k + r)$ is a small positive number modulo one. Apart from $M_k \geq k$ and the monotonicity of $k \mapsto M_k$ (used in Lemma 2(i),(vi)), the proof uses the specific value $M_k = 2^{4^k}$ only in Proposition 5. Given positive $\varepsilon_n \to 0$, choosing M_k nondecreasing with $M_k \geq k$ and $\varepsilon_n \leq 1/(4 \cdot 4^k)$ for all $n \geq M_k$ yields, by the same argument, $m(n+1) \geq (1+\varepsilon_n)m(n)$ for all large n. This recovers an eventual, density-only form of the theorem of [1]; it is not part of the formal verification.

4. The defect and a corrected statement

The statement is a special case of a known theorem. Take $\varepsilon_n = 1/\log(n+2)$, which is positive and tends to 0, in the theorem of Ajtai, Havas and Komlós as recorded in [3, p. 45] (with indices shifted to start at 0). The resulting (m_n) has positive terms, is strictly increasing because every ratio exceeds 1, and satisfies (1) with $c = 1/2$: for $n \geq 2$ we have $n + 2 \leq n^2$, so $\log(n+2) \leq 2\log n$ and $1/\log(n+2) \geq 1/(2\log n)$. Uniform distribution modulo one implies density. Hence `problem_10_6_variant_1` follows from a result published in 1983 and recorded in the book from which Problem 10.6 is taken.

First, the tag `research open` and the docstring sentences saying that the ratio reading "remains open" are incorrect. Second, in both natural cases of an eventual ratio floor $m_{n+1}/m_n \geq 1 + \varepsilon_n$ the answer is already known: floors with $\varepsilon_n \to 0$ are attainable by [1], while a floor $\varepsilon_n \geq \lambda - 1 > 0$ makes the sequence eventually lacunary, and then density fails for some irrational ξ by Pollington and de Mathan [10, 7]. Since Bugeaud states the theorem of [1] before posing Problem 10.6, the problem presumably asks for something that a bare existence statement with a ratio condition does not capture, such as an explicit or natural sequence. We do not attempt to formalize that.

Proposed correction. We propose (1) changing the attribute of `problem_10_6_variant_1` to `@[category research solved, AMS 11]`, replacing "and it is the one that remains open" in its docstring (and the corresponding sentence in the module docstring) by a reference to [1] and [3, §2.8]; and (2) adding the theorem of [1], in the density form that suffices here, as a solved statement:

```
/-- Ajtai-Havas-Komlós [AHK83], as recorded in [Bug12, §2.8], in the weaker form of
density modulo one (they obtain uniform distribution). -/
@[category research solved, AMS 11]
theorem problem_10_6_variants.ajtai_havas_komlos (ε : ℕ → ℝ) (hε : ∀ n, 0 < ε n)
    (hlim : Tendsto ε atTop (nhds 0)) :
    ∃ m : ℕ → ℕ, (∀ n, 0 < m n) ∧ (∀ n, 1 + ε n < (m (n + 1) : ℝ) / m n) ∧
    ∀ ξ : ℝ, Irrational ξ →
      Dense (Set.range fun n => (↑(ξ * m n) : AddCircle (1 : ℝ))) := by
  sorry
```

In a scratch file outside the repository, with the pinned versions of Section 5, we checked that this statement elaborates in `namespace Bugeaud06` and that, taken as a hypothesis, it implies `problem_10_6_variant_1` by the argument above, using only the three standard axioms. The math0 result does not depend on it.

5. Machine verification

File and declarations. The proof is the file `lean/Math0/Books/Bugeaud06_problem_10_6_variant_1.lean` of the math0 repository (`https://github.com/Foundation-0/math0`). It imports only the upstream file `Problem10_6.lean` and the audit commands `lean/Math0/Audit.lean`. In `namespace Math0.Bugeaud06` it defines the parameters $K_k, M_k, D_k, W_k, A_k, L_k$ of Section 3 as `bK`, `bM`, `bD`, `bW`, `bA`, `bL`, the block element `elt k r j` $= A_k + rW_k + (D_k + r)j$, the set `S` exactly as in Section 3, and `seq := Nat.nth (· ∈ S)`, the increasing enumeration of S. The lemmas correspond to Section 3 as follows: Lemma 2 is `bK_le_bD`, `bA_eq`, `bA_eq'`, `two_mul_bA`, `key_bound`, `elt_ge`, `elt_add_le`, `within_gap` and `bL_lt`; Lemma 3 is `gap_block`, `gap_filler` and `S_cases`; Lemma 4 is `S_infinite`, `seq_strictMono`, `seq_mem`, `range_seq`, `seq_le_pow` and `seq_pos`; Proposition 5 is `sublacunary` (which proves `IsGenuinelySublacunary seq` with $c = 1/8$, using Mathlib's bound $\log 2 > 0.6931471803$); Lemma 6 is `step_pos` and `step_any`; Proposition 7 is `dense_aux`; and the final projection to $\mathbb{R}/\mathbb{Z}$ is `dense_seq`. The file ends with

```
theorem problem_10_6_variant_1 : type_of% @Bugeaud06.problem_10_6_variant_1 :=
  ⟨seq, seq_strictMono, sublacunary, dense_seq⟩

end Math0.Bugeaud06

#math0_proves Math0.Bugeaud06.problem_10_6_variant_1 Bugeaud06.problem_10_6_variant_1
```

The file contains no `sorry`, `native_decide`, `set_option` or new axioms, and it compiles in about 15 seconds.

Audit. The command `#math0_proves`, defined in `lean/Math0/Audit.lean`, fails unless (1) the two declarations have the same universe parameters, (2) the type of `Math0.Bugeaud06.problem_10_6_variant_1` is syntactically identical to the type of the upstream `Bugeaud06.problem_10_6_variant_1`, up to metadata annotations only (no unfolding and no definitional-equality check), and (3) our theorem depends on no axioms other than `propext`, `Classical.choice` and `Quot.sound`, which rules out `sorryAx`, the axioms behind `native_decide` and custom axioms. On this file it reports

```
math0 audit: 'Math0.Bugeaud06.problem_10_6_variant_1' depends on axioms [Classical.choice, Quot.
    sound, propext]
math0 statement check: 'Math0.Bugeaud06.problem_10_6_variant_1' proves 'Bugeaud06.
    problem_10_6_variant_1'
```

Toolchain and reproduction. We use Lean 4 [9] `v4.33.1`, Mathlib [8] `v4.33.1` and formal-conjectures [4] at commit `c77fdd446d23539302aa73bcfae4a9fa962b8382`, all pinned in `lean/`. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. A single file can be checked with `lake env lean` from `lean/`.

References

[1] M. Ajtai, I. Havas, and J. Komlós, *Every group admits a bad topology*, in: Studies in Pure Mathematics (to the memory of P. Turán), Birkhäuser, Basel, 1983, 21–34.

[2] M. D. Boshernitzan, *Homogeneously distributed sequences and Poincaré sequences of integers of sublacunary growth*, Monatsh. Math. **96** (1983), 173–181.

[3] Y. Bugeaud, *Distribution Modulo One and Diophantine Approximation*, Cambridge Tracts in Mathematics **193**, Cambridge University Press, Cambridge, 2012.

[4] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An open and evolving benchmark for verified discovery in mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[5] H. Furstenberg, *Disjointness in ergodic theory, minimal sets, and a problem in Diophantine approximation*, Math. Systems Theory **1** (1967), 1–49.

[6] X. Gao, Y. Mooroogen, and C. H. Yip, *On an Erdős similarity problem in the large*, Bull. Lond. Math. Soc. **57** (2025), no. 6, 1801–1818; arXiv:2311.06727 (numbering cited from arXiv v3).

[7] B. de Mathan, *Numbers contravening a condition in density modulo* 1, Acta Math. Acad. Sci. Hungar. **36** (1980), 237–241.

[8] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[9] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[10] A. D. Pollington, *On the density of sequence* $\{n_k\xi\}$, Illinois J. Math. **23** (1979), 511–515.